# FNU TUSHAR

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## EXPERIENCE

**Instructor and Grader, Advanced Database Systems**, New York University — September 2026 - December 2026
New York, NY

- Led technical sessions for graduate students on AQuery, q/kdb+, and ReproZip, covering query execution, time-series data processing, and reproducible database experiments.
- Supported students with database-system assignments and projects involving transaction processing, performance tuning, indexing, and distributed systems.

**Independent Software Consultant**, Confidential Fintech Client — January 2026 - July 2026
Abu Dhabi, UAE

- Developed end-to-end marketplaces for a platform for the financialization and trading of real-world assets.
- Developed custom reinforcement-learning environments in Harbor, including task definitions, reward logic, and evaluation workflows for training and evaluating OpenAI and Anthropic models.

**Software Developer**, Qdata, Sirius Holdings — April 2025 - May 2026
Abu Dhabi, UAE

- Built a CNF generator and SHA-256 implementation from scratch and optimized them for distributed execution.
- Designed a CORBA/OmniORB manager-worker system and containerized workloads on AWS ECS. Used Terraform to provision 1,000+ concurrent workers.
- Developed MT5 trading dashboards and automated multi-leg position-management systems with configurable execution and exit strategies.

**Backend and Data Engineer**, Zator — December 2024 - March 2025

- Developed data streaming pipelines in Go and Rust to ingest JSON and Protobuf data from APIs and Solana RPC.
- Built Kafka streaming workflows with custom producers and consumers for real-time data filtering and processing.

**Young Research Fellow**, IIIT Hyderabad IHub — July 2024 - November 2024

- Built a real-time IoT system to collect sensor data with MQTT, Fastify, and SQLite and display machine learning results in interactive visualizations.
- Researched and implemented on-device ML inference in React Native using a custom ML Kit wrapper, react-native-fast-tflite for TensorFlow Lite, and onnx-runtime-react-native for ONNX.

## EDUCATION

**Master of Science in Computer Science**, New York University — September 2026 - May 2028

**Bachelor of Engineering in Information Science and Engineering**
National Institute of Engineering, Mysore — January 2021 - August 2024

## SKILLS

**Programming languages:** Go, Rust, TypeScript
**Frameworks and libraries:** React Native, Fastify, ML Kit, TensorFlow Lite, ONNX Runtime
**Databases:** SQLite, Postgres, ClickHouse, q/kdb+, TigerBeetle
**Tools and software:** Kafka, Docker, Terraform, MT5, CORBA, OmniORB, Cargo
**Cloud platforms:** AWS: EC2, ECS, ECR, S3, IAM, VPC, VPN, Lambda, CloudWatch. Cloudflare: R2, Workers. Modal.

## PROJECTS

**Monkey Compiler**

- Built an interpreter in Go with no external dependencies, supporting integers, booleans, functions, closures, conditionals, and lexical scoping.

**Market Maker Simulator**

- Built an asynchronous ETH/USDC market-making simulator in Rust. Ingested Binance WebSocket and Jupiter/CoW Protocol API prices concurrently and maintained the latest cross-venue market state through Tokio channels.
- Designed median-based quoting and probabilistic execution models with 2-second trading ticks, automatic WebSocket reconnection, dynamic fill probabilities, and real-time PnL and notional tracking.